INCH-POUND
MIL-C-24706(SH)
13 July 1988

MILITARY SPECIFICATION

COUPLING, PIPE, HEAT-RECOVERABLE

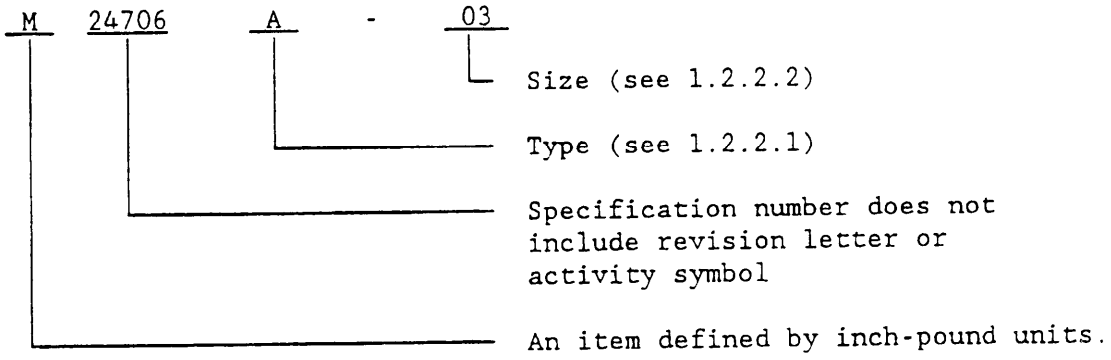
This specification is approved for use within the Naval Sea Systems Command, Department of the Navy, and is available for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers heat-recoverable couplings for permanent pipe-to-pipe, pipe-to-fitting, tube-to-tube, and tube-to-fitting connections in fluid service.

1.2 Classification. Couplings shall be classified by standard part numbers, as specified in 1.2.1 and 1.2.2 (see 6.2.1).

1.2.1 Heat-recoverable coupling part numbers. The standard part numbers for the coupling shall be designated as follows:



1.2.2 Heat-recoverable coupling type and size designators.

1.2.2.1 Types.

Type I - Monolithic coupling
Type II - Composite coupling

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: Commander, Naval Sea Systems Command, SEA 55Z3, Department of the Navy, Washington, DC 20362-5101 by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

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1.2.2.2 Sizes (types I and II).

Nominal pipe size (nps)		Tube outside diameter (od) (inches)	
-01:	1/8	-09:	1/4
-02:	1/4	-10:	3/8
-03:	3/8	-11:	1/2
-04:	1/2	-12:	5/8 (type I only)
-05:	3/4	-13:	3/4
-06:	1	-14:	1
-07:	1-1/4	-15:	1-1/4
-08:	1-1/2	-16:	1-1/2

2. APPLICABLE DOCUMENTS

2.1 Government documents.

2.1.1 Specifications and standards. The following specifications and standards form a part of this specification to the extent specified herein. Unless otherwise specified, the issues of these documents shall be those listed in the issue of the Department of Defense Index of Specifications and Standards (DoDISS) and supplement thereto, cited in the solicitation.

SPECIFICATIONS

MILITARY

MIL-S-901 - Shock Tests, H.I. (High-Impact); Shipboard Machinery, Equipment and Systems, Requirements for.
MIL-C-15726 - Copper-Nickel Alloy, Rod, Flat Products (Flat Wire, Strip, Sheet, Bar, and Plate) and Forgings.
MIL-T-16420 - Tube, Copper-Nickel Alloy, Seamless and Welded (Copper Alloy Numbers 706 and 715).
MIL-T-24107 - Tube, Copper, Seamless (Copper Alloy Numbers C10100, C10200, C10300, C10800, C12000, C12200 and C14200).
MIL-P-24691 - Pipe and Tube, Carbon, Alloy and Stainless Steel, Seamless and Welded, General Specification for.
MIL-P-24691/1 - Pipe and Tube, Carbon Steel, Seamless.
MIL-P-24691/3 - Pipe and Tube, Corrosion-Resistant, Stainless Steel, Seamless or Welded.

STANDARDS

FEDERAL

FED-STD-151 - Metals; Test Methods.

MILITARY

MIL-STD-105 - Sampling Procedures and Tables for Inspection by Attributes.
MIL-STD-109 - Quality Assurance Terms and Definitions.

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MILITARY (Continued)

MIL-STD-129 - Marking for Shipment and Storage.
MIL-STD-167-1 - Mechanical Vibrations of Shipboard Equipment (Type I - Environmental and Type II - Internally Excited).
MIL-STD-792 - Identification Marking Requirements for Special Purpose Components.

(Copies of specifications and standards required by contractors in connection with specific acquisition functions should be obtained from the contracting activity or as directed by the contracting activity.)

2.2 Other publications. The following documents form a part of this specification to the extent specified herein. Unless otherwise specified, the issues of the documents which are DoD adopted shall be those listed in the issue of the DoDISS specified in the solicitation. Unless otherwise specified, the issues of documents not listed in the DoDISS shall be the issue of the non-government documents which is current on the date of the solicitation.

AMERICAN NATIONAL STANDARDS INSTITUTE (ANSI)

B16.9 - Factory-Made Wrought Steel Buttwelding Fittings. (DoD adopted)
B46.1 - Surface Texture (Surface Roughness, Waviness and Lay). (DoD adopted)

(Application for copies should be addressed to the American National Standards Institute, 1430 Broadway, New York, NY 10018.)

AMERICAN SOCIETY OF MECHANICAL ENGINEERS (ASME)

Boiler and Pressure Vessel Codes - Rules for Construction of Nuclear Power Plant Components, Section III, Division 1, Appendix 1.

(Application for copies should be addressed to the American Society of Mechanical Engineers, 345 East 47th Street, New York, NY 10017.)

AMERICAN SOCIETY FOR TESTING AND MATERIALS (ASTM)

B 466 - Standard Specification for Seamless Copper-Nickel Pipe and Tube.
E 8 - Standard Test Methods of Tension Testing of Metallic Materials. (DoD adopted)
E 18 - Standard Test Methods for Rockwell Hardness and Rockwell Superficial Hardness of Metallic Materials. (DoD adopted)
E 112 - Standard Methods for Determining Average Grain Size. (DoD adopted)

(Application for copies should be addressed to the American Society for Testing and Materials, 1916 Race Street, Philadelphia, PA 19103.)

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(Nongovernment standards and other publications are normally available from the organizations which prepare or which distribute the documents. These documents also may be available in or through libraries or other informational services.)

2.3 Order of precedence. In the event of a conflict between the text of this specification and the references cited herein (except for associated detail specifications, specification sheets or MS standards), the text of this specification shall take precedence. Nothing in this specification, however, shall supersede applicable laws and regulations unless a specific exemption has been obtained.

3. REQUIREMENTS

3.1 First article. When specified in the contract or purchase order, a sample shall be subjected to first article inspection (see 4.3 and 6.3).

3.2 Material.

3.2.1 Heat-recoverable material.

3.2.1.1 Composition. The heat-recoverable alloy shall have a composition (see 4.7.2.1) within the range specified below so that the properties of the resulting alloy meet the metallurgical and mechanical requirements specified in 3.2.1.2 through 3.2.1.6. When specified in the contract or purchase order, certification shall be prepared (see 6.2.2).

Element	Weight (percent)
Titanium	43-46
Iron	2-5
Other	1 maximum
Nickel	Balance

3.2.1.2 Grain size. The grain size number shall be numerically 4 minimum (see 4.7.2.2).

3.2.1.3 Nonmetallic inclusions and porosity. Nonmetallic inclusions and porosity shall not exceed 0.001 inch in diameter, and a maximum of 2 percent of the field area shall be occupied by inclusions and porosity which are 0.0005 to 0.001 inch in diameter (see 4.7.2.3).

3.2.1.4 Tensile properties. At ambient temperature, the tensile properties shall be as follows (see 4.7.2.4):

Property	Minimum requirement
Yield strength, 0.2 percent offset	50,000 pounds per square inch (lb/in^2)
Ultimate tensile strength	100,000 lb/in^2
Ultimate elongation	12 percent

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3.2.1.5 Hardness. The hardness shall be Rockwell A-56 minimum (see 4.7.2.5).

3.2.1.6 Dimensional recovery. The permanent dimensional deformation shall be plus or minus 0.3 percent maximum (see 4.7.2.6).

3.2.2 Liner material - (type II couplings only).

3.2.2.1 Composition. The liner material shall be 70-30 copper-nickel (hard temper) in accordance with MIL-C-15726 or ASTM B 466 (see 4.5.3).

3.2.2.2 Hardness. The liner material shall be a minimum hardness of Rockwell B-55 (see 4.7.2.5).

3.3 Construction.

3.3.1 Working pressure and temperature. The maximum allowable working pressure for type I or type II couplings shall be 6000 lb/in^2 from minus 65 to plus 575 degrees Fahrenheit (°F).

3.3.1.1 Type I - coupling. The type I monolithic (one-piece) coupling shall be manufactured from a heat-recoverable material (see 3.2.1). The inside diameter (id) shall contain annular sealing and gripping lands which shall produce a high performance joint when installed (see figure 1).

3.3.1.2 Type II - coupling. The type II coupling shall be composed of two parts: the driver, which shall be made from a heat-recoverable material (see 3.2.1), and a 70-30 copper-nickel liner (see 3.2.2), which shall isolate the heat-recoverable driver material from the system fluid (see figure 2). The inside diameter of the liner shall contain annular sealing and gripping lands. The liner shall be retained in the driver so that the coupling is shipped and installed as an assembly. The coupling shall be so constructed that, when installed, the sealing and gripping lands produce a high performance joint.

3.3.2 Dimensions.

3.3.2.1 Coupling dimensions. Dimensions of the type I or type II coupling shall be in accordance with table I or table II as applicable (see 4.7.1).

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TABLE I. Type I coupling construction details.[1/]

(See figure 1)

	Description	Nominal od	"A" diameter		"B" diameter	"C" length	"D" wall thickness
			As-fabricated (nominal)[2/3/]	As-expanded (minimum)[4/]	As-fabricated (nominal)[2/3/]	As-fabricated (nominal)[2/3/]	As-fabricated (minimum)[3/]
Pipe nps	1/8	0.405	0.381	0.408	0.676	1.02	0.132
	1/4	.540	.508	.544	.861	1.37	.165
	3/8	.675	.636	.681	1.055	1.70	.191
	1/2	.840	.793	.848	1.263	2.12	.214
	3/4	1.050	.991	1.062	1.537	2.65	.249
	1	1.315	1.241	1.330	1.863	3.33	.283
	1-1/4	1.660	1.566	1.674	2.243	4.19	.307
	1-1/2	1.900	1.794	1.916	2.525	4.80	.330
Tube od	1/4	0.250	0.235	0.252	0.434	0.65	0.095
	3/8	.375	.353	.378	.622	.95	.120
	1/2	.500	.473	.506	.811	1.26	.157
	5/8	.625	.592	.632	.987	1.58	.180
	3/4	.750	.709	.758	1.167	1.89	.210
	1	1.000	.945	1.009	1.481	2.52	.245
	1-1/4	1.250	1.181	1.259	1.788	3.15	.257
	1-1/2	1.500	1.416	1.509	2.076	3.79	.300

1/ All dimensions are in inches.
2/ See 3.3.2.3 for tolerances.
3/ As-fabricated dimensions are the machined dimensions prior to expansion.
4/ As-expanded dimensions are after the coupling has been cooled and expanded.

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TABLE II. Type II coupling construction details.[1/]

(See figure 2)

	Description	Nominal od	"A" diameter (liner) Liquid nitrogen temperature (minimum)	"B" diameter As-fabricated or as-recovered (nominal)[2/]	"C" length As-fabricated (nominal)[2/]	"D" diameter (driver) As-fabricated (maximum)	"D" diameter (driver) As-expanded (minimum)	"E" wall thickness As-fabricated (minimum)
Pipe nps	1/8	0.405	0.408	0.740	1.02	0.453	0.476	0.020
	1/4	.540	.543	.918	1.37	.593	.626	.025
	3/8	.675	.680	1.102	1.70	.735	.775	.028
	1/2	.840	.845	1.338	2.12	.930	.978	.031
	3/4	1.050	1.056	1.618	2.65	1.145	1.205	.040
	1	1.315	1.321	1.961	3.33	1.440	1.518	.065
	1-1/4	1.660	1.670	2.528	4.19	1.796	1.894	.070
	1-1/2	1.900	1.910	3.125	4.80	2.062	2.176	.090
Tube od	1/4	0.250	0.2505	0.505	0.65	0.293	0.308	0.018
	3/8	.375	.378	.680	.95	.419	.440	.019
	1/2	.500	.504	.867	1.28	.557	.588	.025
	3/4	.750	.756	1.210	1.89	.816	.860	.030
	1	1.000	1.006	1.580	2.52	1.099	1.157	.040
	1-1/4	1.250	1.257	1.910	3.15	1.380	1.453	.062
	1-1/2	1.500	1.510	2.260	3.79	1.634	1.720	.067

1/ All dimensions are in inches.
2/ See 3.3.2.3 for tolerances.

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3.3.2.2 Recovered od. When a type I coupling or type II heat recoverable driver is recovered on dry ice (approximately minus 108°F), the od shall be not larger than the as-fabricated od (see 4.7.3.3).

3.3.2.3 Tolerances. Dimensions shall not vary from the nominal values listed in table I and table II by more than plus or minus 0.125 inch for dimensions B and C or by more than 0.010 inch for all other dimensions.

3.3.2.4 Internal machined surfaces. Internal machined surfaces shall be 125 roughness height rating maximum as specified in ANSI B46.1 (see 4.7.1).

3.4 Performance.

3.4.1 General requirements. A test specimen shall consist of two pieces of pipe or tube joined with a coupling. The specimen shall be approximately 25 inches long with the coupling located in the center of the specimen. No means of joining the two pieces of pipe other than the normal contraction of the coupling caused by recovery of the coupling driver shall be permitted. The od of the pipe or tube shall be within the range specified in table III; however, the cut pipe ends may be deburred for easier assembly. The wall thickness shall be the minimum or maximum specified in table III. The outside surface within the coupling area shall be free of longitudinal or helical scratches, irregularities, or foreign matter and shall be convex at all points with no discontinuity in curvature. After the test specimen is assembled, the pipe or tube ends shall not be displaced from the center of the coupling by greater than the amount specified in table III. Maximum pipe end displacement shall be obtained by measuring the test specimen prior to and after assembly.

TABLE III. Dimensions of pipe and assembly.1/

Pipe nps	Outside diameter				Maximum pipe end displacement from coupling center
	Nominal	Minimum	Maximum Type I	Maximum Type II	
1/8	0.405	0.399	0.407	0.407	0.062
1/4	.540	.532	.543	.542	.068
3/8	.675	.666	.680	.679	.084
1/2	.840	.830	.847	.844	.105
3/4	1.050	1.038	1.061	1.055	.131
1	1.315	1.300	1.329	1.320	.164
1-1/4	1.66	1.640	1.673	1.668	.207
1-1/2	1.900	1.880	1.916	1.910	.237
Tube od					
1/4	0.250	0.245	0.2505	0.250	0.062
3/8	.375	.369	.377	.376	.062
1/2	.500	.494	.505	.504	.062
5/8	.625	.619	.631	.630	.078
3/4	.750	.743	.757	.755	.093
1	1.000	.990	1.008	1.005	.125
1-1/4	1.250	1.238	1.258	1.255	.156
1-1/2	1.500	1.485	1.508	1.505	.188

See footnote at end of table.

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TABLE III. Dimensions of pipe and assembly. 1/ - Continued

Type I
Pipe wall thickness

Size	Nominal od	Carbon steel MIL-P-24691/1		CRES MIL-P-24691/3		70/30 Cu-Ni MIL-T-16420	
		Min	Max	Min	Max	Min	Max
Pipe (nps)							
1/8	0.405	0.068	0.095	0.068	0.095	0.058	0.095
1/4	.540	.088	.119	.088	.140	.042	.120
3/8	.675	.091	.126	.091	.126	.049	.148
1/2	.840	.109	.147	.109	.188	.058	.203
3/4	1.050	.113	.154	.113	.219	.083	.238
1	1.315	.133	.179	.133	.250	.095	.300
1-1/4	1.660	.140	.191	.140	.250	.095	.380
1-1/2	1.900	.145	.200	.145	.281	.109	.425
Tube (od)							
1/4	0.250	0.035	0.065	0.035	0.058	0.035	0.058
3/8	.375	.049	.083	.049	.083	.049	.083
1/2	.500	.065	.120	.065	.120	.042	.120
5/8	.625	.083	.126	.083	.126	-	-
3/4	.750	.095	.147	.095	.156	.058	.165
1	1.000	.109	.156	.109	.219	.072	.220
1-1/4	1.250	.134	.188	.134	.250	.095	.284
1-1/2	1.500	.134	.200	.134	.250	.109	.340

Type II
Pipe wall thickness

Size	Nominal od	Copper MIL-T-24107		90/10 Cu/Ni MIL-T-16420		70/30 Cu-Ni MIL-T-16420	
		Min	Max	Min	Max	Min	Max
Pipe (nps)							
1/8	0.405	0.065	0.109	0.035	-	-	-
1/4	.540	.065	.134	.065	-	0.065	0.120
3/8	.675	.065	.165	.065	-	.065	.148
1/2	.840	.065	.220	.065	-	.065	.203
3/4	1.050	.065	.263	.065	-	.065	.238
1	1.315	.065	.340	.065	-	.065	.300
1-1/4	1.660	.065	.340	.072	-	.072	.340
1-1/2	1.900	.065	.380	.072	-	.072	.380

See footnote at end of table.

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TABLE III. Dimensions of pipe and assembly.1/ - Continued

Type II
Pipe wall thickness

Size	Nominal od	Copper MIL-T-24107		90/10 Cu/Ni MIL-T-16420		70/30 Cu-Ni MIL-T-16420	
		Min	Max	Min	Max	Min	Max
Tube (od)							
1/4	0.250	0.032	0.065	0.035	0.065	0.035	0.058
3/8	.375	.032	.065	.035	-	.049	.083
1/2	.500	.032	.065	.035	-	.042	.120
5/8	.625	-	-	-	-	-	-
3/4	.750	.065	.165	.065	-	.058	.165
1	1.000	.065	.263	.065	-	.072	.220
1-1/4	1.250	.065	.263	.065	-	.095	.284
1-1/2	1.500	.065	.340	.072	-	.109	.340

1/ All dimensions are in inches.

3.4.1.1 Hydraulic fluid. For the purpose of testing (see 4.7.4) hydraulic fluid shall be hydraulic oil or water.

3.4.1.2 Test temperatures. Unless otherwise specified herein, all tests shall be performed at an ambient temperature of 70 ± 10°F.

3.4.1.3 Test working pressure. The test working pressures shall be as follows:

Pipe material	Schedule	Test working pressure (lb/in^2)
Stainless steel/ carbon steel	40	1500
	80	3000
	160	6000
	other	1500
Copper-nickel	--	Maximum working pressure per pipe specification (for wall thickness under test)
Copper	--	Allowable working pressure per pipe specification (for thickness under test)

3.4.2 Gas leak. The test specimens shall show no evidence of leakage (see 4.7.4.2).

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WITH OR WITHOUT CONTINUITY SWITCH

3.4.3 Hydrostatic proof pressure. The test specimen shall show no evidence of leakage at the coupling and the coupling shall show no bulges or cracks (see 4.7.4.3).

3.4.4 Hydrostatic burst pressure. The minimum burst pressure shall be pipe burst, gross yielding of the pipe or the minimum calculated burst pressure of the pipe, as specified in ANSI B16.9, whichever occurs first (see 4.7.4.4). Gross yielding of the pipe shall be defined as a 25 percent increase in the od of the pipe.

3.4.5 Hydrostatic impulse followed by flexural fatigue. A constant bending load, which results in a pipe bending stress of two times the value specified in table IV, shall be applied to the specimen. The specimen shall then be impulse tested for 1,000,000 cycles (see 4.7.4.5). The same specimen shall then be flexural fatigue tested in accordance with 3.4.6 for 80,000 cycles. The specimen shall show no evidence of leakage during or at the completion of testing.

TABLE IV. Pipe bending stress.

Material	Test stress value (klb/in^2)	Modulus of elasticity (lb/in^2 by 10^6)
Carbon steel	31	29.0
CRES (type 304 and 316)	30	28.5
CRES (type 304L and 316L)	25	28.5
Cu-Ni (70-30)	22	22.0
Cu-Ni (90-10)	15	18.0
Copper	13.5 1/	17.0

1/ 1.5 times the allowable stress value specified in ASME Boiler and Pressure Vessel Code, section 3, appendix 1.

3.4.6 Pressurized planar flexural fatigue. The test specimen shall withstand a minimum of 80,000 cycles of planar flexural bending without leakage of the coupling or connection, while pressurized at 1.2 to 1.3 times the test working pressure of the pipe or tube as specified in 3.4.1.3 (see 4.7.4.6). The maximum deflection during flexural fatigue testing shall produce a minimum strain equivalent to either plus two times the stress levels specified in table IV (see figure 3, method 1) or plus one and minus one times the stress levels specified in table IV (see figure 3, method 2). The specimen shall show no evidence of leakage during or at the completion of testing. After fatigue testing the test specimen shall be proof tested (see 4.7.4.3) and shall meet the requirements of 3.4.3.

3.4.7 Tensile pull-out strength. The minimum tensile strength of the coupling-to-pipe test specimen shall be as follows:

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Minimum tensile strength = $A_p \times S_y$

Where: A_p = actual cross-sectional area of pipe (square inches)
S_y = minimum specified yield strength of pipe at test temperature in accordance with ASME Boiler and Pressure Vessel Code, section 3, appendix 1.

Tensile strength shall be the highest tensile load value observed (see 4.7.4.7).

3.4.8 Shock. The test specimen shall be in accordance with MIL-S-901 for grade A, class I, hull-mounted equipment (see 4.7.4.8).

3.4.9 Vibration. The test specimen shall meet the requirements for type I vibrations in accordance with MIL-STD-167-1 (see 4.7.4.9).

3.4.10 Thermal cycling (type I couplings only). Test specimens shall be subjected to three elevated temperature thermal cycles (see 4.7.4.10). After each cycle, the specimen shall meet the requirements of 3.4.2. The test specimen shall then be tensile tested (see 4.7.4.7) and shall meet the requirements of 3.4.7. Separate test specimens shall be subjected to three low temperature thermal cycles (see 4.7.4.10.1). The test specimens shall be examined after each cycle and shall show no evidence of leakage. The test specimens shall then be tensile tested at ambient temperature (see 4.7.4.7) and shall meet the requirements of 3.4.7.

3.4.11 Elevated temperature soak (type I couplings only). The test specimen shall meet the requirements specified in 3.4.4 after being subjected to elevated temperature soak (see 4.7.4.11).

3.5 Identification marking. Each coupling shall be marked legibly and permanently in accordance with MIL-STD-792 and shall include the following information:

(a) Manufacturer's name or trademark.
(b) Specification part number (see 1.2).
(c) Lot number.
(d) Pressure rating.

3.6 Workmanship. Surfaces shall be free of burrs, cracks, laps or seams which are visible to the unaided eye and which would affect the suitability for the intended service. Sealing surfaces shall be free of any continuous flaws which would constitute a leak path.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the contractor is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or purchase order, the contractor may use his own or any other

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facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.1.1 Responsibility for compliance. All items must meet all requirements of sections 3 and 5. The inspection set forth in this specification shall become a part of the contractor's overall inspection system or quality program. The absence of any inspection requirements in the specification shall not relieve the contractor of the responsibility of assuring that all products or supplies submitted to the Government for acceptance comply with all requirements of the contract. Sampling in quality conformance does not authorize submission of known defective material, either indicated or actual, nor does it commit the Government to acceptance of defective material.

4.1.2 Inspection system. When specified in the contract or order an inspection system program plan shall be prepared (see 6.2.2).

4.2 Classification of inspections. The inspection requirements specified herein are classified as follows:

(a) First article inspection (see 4.3).
(b) Quality conformance inspection (see 4.4).
(c) Process control inspection (see 4.5).

4.3 First article inspection. First article inspection shall consist of the examinations and tests specified in table V.

TABLE V. First article inspection and tests.

Test	Requirement	Method	Test samples per pipe material					
			Minimum diameter coupling		Intermediate diameter coupling		Maximum diameter coupling	
			Min wall pipe	Max wall pipe	Min wall pipe	Max wall pipe	Min wall pipe	Max wall pipe
Gas leak	3.4.2	4.7.4.2	20	22	10	8	4	8
Hydrostatic proof	3.4.3	4.7.4.3	20	22	10	8	4	8
Hydrostatic burst	3.4.4	4.7.4.4						
Ambient temp.			3	3		1		1
Low temp. 1/			3	3				
High temp. 1/			3	3	1	1	1	1

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TABLE V. First article inspection and tests. - Continued

Test	Requirement	Method	Test samples per pipe material					
			Minimum diameter coupling		Intermediate diameter coupling		Maximum diameter coupling	
			Min wall pipe	Max wall pipe	Min wall pipe	Max wall pipe	Min wall pipe	Max wall pipe
Impulse followed by flexural fatigue	3.4.5	4.7.4.5	1	1	1	1	1	1
Pressurized planar flexural fatigue	3.4.6	4.7.4.6	1	1		1	1	1
Tensile pull-out strength	3.4.7	4.7.4.7						
Ambient temp.			3	3		1		1
Low temp.1/			3	3				
High temp.1/			3	3	1	1	1	1
Shock	3.4.8	4.7.4.8		1		1		1
Vibration	3.4.9	4.7.4.9		1		1		1
Thermal cycling (type I only)2/	3.4.10	4.7.4.10						
Elevated temp.					3			
Low temperature					3			
Elevated temp. soak (type I only)2/	3.4.11	4.7.4.11			1			

1/ See 4.7.4.4 and 4.7.4.7.
2/ Required for first article inspection of type I couplings on stainless steel pipe or carbon steel pipe only.

4.3.1 Specimen sizes. The number of test specimens required is listed in table V. For any size range, test shall be conducted on the smallest diameter coupling, an intermediate diameter coupling, and the largest diameter coupling of each type. Tests shall be conducted on the minimum wall thickness pipe and maximum wall thickness pipe for each coupling size tested (see table III). Use of each coupling type shall be limited to the range of diameters tested and does not extend to smaller or larger diameters. Type I and type II couplings shall be tested separately. Tests conducted on one type do not apply to the other type.

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First article inspection of a coupling type requires completion of all specified tests using all pipe materials applicable to that coupling type. Unless otherwise specified, test assemblies shall be subjected to a gas leak test in accordance with 4.7.4.2 and hydrostatic proof test in accordance with 4.7.4.3 prior to other tests.

4.4 Quality conformance inspection. Quality conformance inspection shall consist of the examinations and tests specified in table VI. When specified in the contract or order, certification data/reports shall be prepared (see 6.2.2). Quality conformance inspection shall be performed on every lot of couplings acquired under this specification. Acceptance quality level (AQL) shall be as specified in table VI, determined in accordance with MIL-STD-105.

TABLE VI. Quality conformance inspection.

Examination or test	Requirement	Test method	AQL level
Expanded id	Table I and II	4.7.3.1	Level S-1, acceptance number 0
Recovered id	Table I and II	4.7.3.2	Level S-1, acceptance number 0
Recovered od 1/	3.3.2.2	4.7.3.3	One coupling or driver per lot of material, acceptance number 0

1/ This test shall be performed on couplings from every lot of material as defined in 4.4.1.5.

4.4.1 Lot for quality conformance inspection. Definitions of inspection terms used herein shall be in accordance with MIL-STD-109. The lot for quality conformance inspection shall be as specified in 4.4.1.1 through 4.4.1.5.

4.4.1.1 Lot of couplings. The inspection lot shall include all type I couplings of the same size manufactured at essentially the same time from the same lot of material and submitted for inspection at one time, or all type II couplings of the same size assembled from the same lot of drivers and offered for inspection at the same time.

4.4.1.2 Lot of type II drivers. A lot of drivers shall consist of all drivers of the same size, manufactured at essentially the same time from the same lot of material and submitted for inspection at the same time.

4.4.1.3 Lot of type II liners. A lot of liners shall consist of all liners of the same size, manufactured at essentially the same time from the same lot of liner material and submitted for inspection at the same time.

4.4.1.4 Lot of type II liner material. A lot of liner material shall consist of bars, tubes, forgings, or other liner material of the same heat, produced at essentially the same time and submitted for inspection at the same time.

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4.4.1.5 Lot of heat-recoverable material. A lot of material shall consist of bars, tubes, or forgings produced from not more than one ingot.

4.4.2 Nonconforming inspection lots. Disposition of inspection lots found unacceptable shall be in accordance with MIL-STD-105.

4.5 Process control inspection. Process control inspection shall consist of the examinations and tests of tables VII and VIII, which are of such a nature that they cannot be performed on the finished product and therefore must be performed at an appropriate stage of manufacture. Process control inspection shall be performed on every lot of material, type I couplings, type II drivers and type II liners acquired under this specification as appropriate. When specified in the contract or order, certification data/report shall be prepared (see 6.2.2).

TABLE VII. Process control inspection for material.

Examination or test	Requirement	Test method	Number of samples from each lot
Heat-recoverable material			
Composition	3.2.1.1	4.7.2.1	1
Grain size	3.2.1.2	4.7.2.2	1
Nonmetallic inclusions and porosity	3.2.1.3	4.7.2.3	1
Yield strength at ambient temperature	3.2.1.4	4.7.2.4	1
Ultimate tensile strength at ambient temperature	3.2.1.4	4.7.2.4	1
Ultimate elongation at ambient temperature	3.2.1.4	4.7.2.4	1
Hardness	3.2.1.5	4.7.2.5	6
Dimensional recovery	3.2.1.6	4.7.2.6	1

TABLE VIII. Process control inspection for type I and type II couplings.

Examination	Requirement	Test method	Inspection level and AQL (see MIL-STD-105)	Stage of manufacturing performed
Diameter A	Table I	4.7.1	Level II, AQL 1.0	As-fabricated
Diameter B	Tables I and II	4.7.1	Level II, AQL 4.0	As-fabricated
Length C	Tables I and II	4.7.1	Level II, AQL 4.0	As-fabricated

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TABLE VIII. Process control inspection for type I and type II couplings. - Continued

Examination	Requirement	Test method	Inspection level and AQL (see MIL-STD-105)	Stage of manufacturing performed
Wall thickness/ diameter D	Tables I and II	4.7.1	Level II, AQL 4.0	As-fabricated
Wall thickness E	Table II	4.7.1	Level II, AQL 4.0	As-fabricated
Workmanship	3.6	4.7.1	Level II, AQL 4.0	As-fabricated
Identification	3.5	4.7.1	Level II, AQL 4.0	As-fabricated

4.5.1 Sampling for process control inspection. Samples shall be selected in accordance with MIL-STD-105 for all process control inspections.

4.5.2 Heat-recoverable material process control inspection. Each lot of heat-recoverable material shall be subjected to the examinations and tests listed in table VII.

4.5.3 Type II liner material process control inspection. Each lot of liner material shall be the composition specified in 3.2.2.1 and shall meet the hardness requirement of 3.2.2.2. When specified in the contract or order, certification shall be prepared (see 6.2.2).

4.5.4 Coupling process control inspection. Each lot of coupling shall be subjected to the examinations and tests specified in table VIII.

4.6 Rejection and retest. Reworked lots shall be resubmitted for reinspection and acceptance in accordance with MIL-STD-105.

4.7 Test methods.

4.7.1 Examination of product. Drivers, liners, and couplings shall be visually and dimensionally inspected to verify conformance to the requirements of this specification using standard inspection equipment as appropriate.

4.7.2 Material tests.

4.7.2.1 Composition. The composition of heat-recoverable material shall be determined in accordance with FED-STD-151, method 112.2 and shall meet the requirements of 3.2.1.1.

4.7.2.2 Grain size. The grain size shall be determined in accordance with ASTM E 112. The test specimen shall be a minimum 0.1 square inch section taken parallel to the long axis of the stock from which the couplings are manufactured or a coupling machined from the material.

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4.7.2.3 Nonmetallic inclusions and porosity. The size and quantity of nonmetallic inclusions shall be determined by micrographical observation. The specimen shall be a 0.1 square inch section taken parallel to the long axis of the stock from which the components are manufactured (see 3.2.1.3). The surface of the specimen shall be examined at a magnification of 100X within a field of 3.5 square inches. Three fields shall be selected at random for measurement of the inclusions and porosity.

4.7.2.4 Mechanical test. Yield strength, ultimate tensile strength and ultimate elongation shall be determined in accordance with ASTM E 8 and shall meet the requirements of 3.2.1.4. The test apparatus shall maintain the specified temperatures.

4.7.2.5 Hardness. The hardness shall be determined in accordance with ASTM E 18 and shall meet the requirements of 3.2.1.5 and 3.2.2.2.

4.7.2.6 Dimensional recovery. A specimen of heat recoverable material (for example, an ASTM tensile specimen) shall be mounted in the test apparatus and cooled to minus 320 ± 18°F. The specimen shall be subjected to a uniaxial stress which will give a specimen retained axial strain of 6 ± 1/2 percent when the stress is removed. The stress shall then be removed and the specimen shall be heated to 73 ± 5°F. After recovery, the length of the specimen shall be measured and the permanent deformation calculated as follows:

$$D = 100 \frac{L_1 - L_0}{L_0}$$

Where: D = permanent deformation, percent.
L_0 = original gauge length of specimen (inches) at ambient temperature (75 ± 5°F).
L_1 = final gauge length of specimen (inches) at ambient temperature (75 ± 5°F).

The specimen shall meet the requirements of 3.2.1.6.

4.7.3 Coupling tests.

4.7.3.1 Expanded id. The expanded id of the type I or type II coupling (driver plus liner), dimension A of figures 1 and 2, shall be measured in liquid nitrogen. Measuring devices shall be accurate at liquid nitrogen temperature of minus 320 ± 18°F if they are to be submerged in liquid nitrogen. The coupling shall meet the requirements of table I and table II as applicable.

4.7.3.2 Recovered id. An expanded type I coupling previously measured in accordance with 4.7.3.1 shall be removed from the liquid nitrogen and allowed to freely recover and stabilize at ambient temperature. The recovered id of the coupling shall be measured at ambient temperature and shall meet the requirements of table I. The minimum expanded id of the driver from a type II coupling without the liner shall be measured in liquid nitrogen and shall meet the requirements of table II. The driver shall then be removed from the liquid

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nitrogen and allowed to freely recover and stabilize at ambient temperature. The recovered id of the driver shall then be measured at ambient temperature and shall meet the requirements of table II. Samples of either type I or type II couplings used for this test may be re-expanded and replaced into the lot.

4.7.3.3 Recovered od. The expanded type I coupling or type II driver shall be quickly transferred to dry ice and held at dry ice temperature until the temperature stabilizes. The od shall then be not larger than the maximum as-fabricated diameter specified in table I or table II (see 3.3.2.2). Samples used for this test may be re-expanded and replaced into the lot.

4.7.4 Assembled test specimen tests.

4.7.4.1 Failure. For the purpose of testing, failure shall be defined as fracture of the coupling or leakage of the test specimen. Fracture of the pipe more than 1/2 coupling length removed from the coupling end shall not be deemed as coupling failure, and the pipe may be repaired and the testing resumed.

4.7.4.2 Gas leak. The test specimen shall be connected to a regulated air or nitrogen gas source and immersed in water. The gas pressure shall gradually be applied to reach 1.5 times the rated work pressure or 1000 lb/in^2, whichever is lower, in approximately 30 seconds. This pressure shall be held for 5 minutes, while the test assembly is observed for leakage. The specimen shall meet the requirements of 3.4.2.

4.7.4.3 Hydrostatic proof pressure. The test specimen shall be connected to a regulated hydrostatic source and shall be filled with fluid. Pressure shall be increased to 1.5 times the test working pressure of the pipe and held for 5 minutes. Pressure shall then be reduced and the test specimen examined for leakage. The specimen shall meet the requirements of 3.4.3.

4.7.4.4 Hydrostatic burst pressure. The temperature of the test specimen shall be monitored prior to and during testing. Pressure shall be increased to the burst pressure, or until burst, leakage, or gross yielding of the pipe occurs. The specimen shall meet the requirements specified in 3.4.4. Low temperature tests shall be performed at or below minus 65°F. High temperature tests shall be performed at or above 575°F.

4.7.4.5 Hydrostatic impulse followed by planar flexural fatigue. Impulse testing shall be performed with hydraulic fluid at a rate of 60 $\pm$ 5 cycles per minute with each impulse cycle constituting an excursion from a low-level system charging pressure to surge pressure and a return to charging pressure. The peak surge pressure shall be 1.2 to 1.3 times the test working pressure (see 3.4.1.3) above the system charging pressure. The system charging pressure shall be as low as possible, but not higher than 15 percent of the peak surge pressure. During this impulse test, a constant bending moment shall be applied to the test specimen which shall result in a pipe bending stress of two times the value specified in table IV. The pipe bending stress shall be determined by affixing strain gauges to the pipe wall on the high stress side of the test specimen,

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approximately one coupling length removed from each end of the coupling. The pipe shall then be bent until the strain gauges indicate a strain equivalent to the required pipe outer fiber stress, assuming the modulus of elasticity of the pipe remains constant. Under the required bending load the couplings shall complete 1,000,000 impulse cycles. Subsequent to impulse testing, the same assembly shall be tested in planar flexural fatigue in accordance with 4.7.4.6 and shall meet the requirements of 3.4.5. At the conclusion of the number of planar flexural fatigue test cycles, the test specimen shall again be proof pressure tested in accordance with 4.7.4.3.

4.7.4.6 Pressurized planar flexural fatigue. The test specimen shall be subjected to a reciprocating isoplanar strain at the bending stress and for the test duration. The planar bending stress shall be induced by applying a reciprocating isoplanar displacement to bending arms attached to the test specimen. Test configurations shall be as shown on figure 3. The displacement of the bending arms shall be of a magnitude to subject the assembly to the minimum specified strain. To determine the required arm displacement, strain gauges shall be affixed to the high stress side of the pipe on each side of the coupling at three pipe diameters removed from the coupling. The arm shall then be displaced until the strain is reached which yields the required stress (see 3.4.6), assuming a constant modulus of elasticity. The displacement for testing shall be set at this value. The test specimen shall then be pressurized to 1.2 to 1.3 times the test working pressure (see 3.4.1.3), and planar bending shall commence with the constant displacement determined above at a rate not to exceed 50 cycles per minute until failure or the required number of cycles is obtained. One cycle shall be defined as the bending arm excursion from the initial neutral stress position to the maximum arm displacement and return to the initial position. The specimen shall meet the requirements of 3.4.6. After cycling, the test specimen shall be proof pressure tested in accordance with 4.7.4.3.

4.7.4.7 Tensile pull-out strength. The test specimen shall be strained in a constant strain rate tensile unit. The gripping jaws of the tensile unit shall be at least three pipe diameters removed from the coupling envelope. The test specimen shall be strained at a crosshead separation rate of less than 0.05 inch per minute, and tensile load shall be continuously monitored. Low temperature tests shall be performed at or below minus 65°F. High temperature test shall be performed at or above 575°F. Temperatures shall be monitored during testing. The specimen shall meet the requirements specified in 3.4.7.

4.7.4.8 Shock. The test specimen shall be shock tested in accordance with MIL-S-901 to assure conformance to the requirements specified in 3.4.8. The specimen shall be mounted on a standard lightweight mounting fixture, in accordance with MIL-S-901, between split block type clamps that are rigidly secured to the test fixture. The specimen shall be pressurized to 6000 lb/in^2 throughout the test. There shall be no evidence of damage or leakage.

4.7.4.9 Vibration. The test specimen shall be vibration tested in accordance with MIL-STD-167-1. The specimen shall be mounted between split block type clamps that are rigidly secured to the test fixture. Testing shall meet the requirements specified in 3.4.9.

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4.7.4.10 Thermal cycling. Test specimens shall be subjected to elevated temperature thermal cycling as follows. The test specimen shall be mounted in an insulated container. Steam at 575°F and at saturation pressure shall be admitted to the test specimen and flow maintained until the coupling has stabilized at 575°F for a period of 2 hours. Saturated steam at 212°F shall then be introduced into the test specimen at atmospheric pressure. Steam flow at 212°F shall be maintained until the test specimen has stabilized for a period of 2 hours at 212°F. At the end of the stabilization period, steam at 575°F and at saturation pressure shall be introduced for a period of 2 minutes. At the conclusion of the 2-minute period, the test specimen shall be allowed to return to ambient temperature and shall be pressurized in accordance with 4.7.4.2 and shall be examined for leakage. The above procedure shall be repeated for a total of three cycles. The specimen shall meet the requirements specified in 3.4.10.

4.7.4.10.1 Separate test specimens. Separate test specimens shall be subjected to low temperature thermal cycling as follows. The test specimen shall be connected to a regulated hydraulic fluid pressure source and mounted in an environmental chamber. Fluid shall be admitted to the test specimen and the pressure shall be increased to 1000 lb/in^2 minimum and the ambient temperature raised to 275°F and held at this temperature for a minimum of 2 hours. At the end of the above soak, while still at temperature, the hot test fluid shall be released and replaced with test fluid below minus 65°F within a period of 10 seconds or less. The test pressure during this period shall remain at a minimum of 1000 lb/in^2 and the test specimen shall be checked for leakage. The test specimen shall be stabilized to room temperature and the pressure slowly released. The test specimen shall again be pressurized to 1000 lb/in^2 minimum and the ambient temperature lowered to below minus 65°F and held at that temperature for a minimum of 2 hours. At the end of the above soak, the cold fluid shall be released and replaced with test fluid at 275°F within a period of 10 seconds or less. The test pressure during this period shall remain at a minimum of 1000 lb/in^2 and the assembly shall be checked for leakage. The test specimen shall be stabilized to room temperature, and the pressure released. The above procedure shall be repeated for a total of three cycles. The specimen shall meet the requirements specified in 3.4.10.

4.7.4.11 Elevated temperature soak. The test specimen shall be maintained, unpressurized, at a minimum temperature of 700°F for a minimum of 600 hours. At the completion of the 600 hours the test specimen shall be returned to ambient temperature and shall be burst tested in accordance with 4.7.4.4. The coupling-to-pipe assembly shall meet the requirements specified in 3.4.11.

4.8 Inspection of packaging. Sample packages and packs, and the inspection of the preservation, packing and marking for shipment, stowage and storage shall be in accordance with the requirements of section 5 and the documents specified therein.

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5. PACKAGING

(The packaging requirements specified herein apply only for direct Government acquisition.)

5.1 Preservation, levels A, C, or commercial. Preservation of couplings to maintain the required cryogenic state shall be immersed in nonpressurized liquid nitrogen Dewar flasks or containers. Container liquid volume shall be 13 gallons maximum, shall be vented for atmospheric change, and capped to prevent spillage.

5.2 Packing, levels A, B, C, or commercial. Couplings preserved as specified in 5.1 shall be palletized to facilitate handling and to protect against impact, overturning, or loss of contents from spillage. The method of securing the containers to the pallet shall ensure the integrity of the pallet load.

5.3 Marking. In addition to any special marking required (see 6.2.1), containers and pallet loads shall be marked in accordance with MIL-STD-129.

6. NOTES

6.1 Intended use. Type I couplings are intended for use at a maximum pressure rating of 6000 lb/in^2 and a temperature rating of minus 65 to plus 575°F for water, oil, or gas service. Type I couplings should not be used in seawater service. Type I couplings are to be installed on piping systems fabricated from 70-30 copper-nickel, corrosion resistant steel or carbon steel only. Type II couplings are intended for use at a maximum pressure rating of 6000 lb/in^2 and a temperature rating of minus 65 to plus 575°F for water, oil, gas, and seawater service. Type II couplings should be installed on piping systems fabricated from 70-30 copper-nickel, 90-10 copper-nickel or copper only.

6.2 Ordering data.

6.2.1 Acquisition requirements. Acquisition documents should specify the following:

(a) Title, number, and date of this specification.
(b) Specification part number (1.2.1).
(c) First article inspection, when required (see 3.1).
(d) Special marking, if required (see 5.3).

6.2.2 Data requirements. When this specification is used in an acquisition and data are required to be delivered, the data requirements identified below shall be developed as specified by an approved Data Item Description (DD Form 1664) and delivered in accordance with the approved Contract Data Requirements List (CDRL), incorporated into the contract. When the provisions of DoD FAR Supplement, Part 27, Sub-Part 27.475-1 (DD Form 1423) are invoked and the DD Form 1423 is not used, the data specified below shall be delivered by the contractor in accordance with the contract or purchase order requirements. Deliverable data required by this specification are cited in the following paragraphs.

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Paragraph no.	Data requirement title	Applicable DID no.	Option
3.2.1.1 and 4.5.3	Certificate of compliance	DI-E-2121	----
4.1.2	Inspection system program plan	DI-R-4803	----
4.4 and 4.5	Certification data/report	UDI-A-23264	----

(Data item descriptions related to this specification, and identified in section 6 will be approved and listed as such in DoD 5010.12-L., AMSDL. Copies of data item descriptions required by the contractors in connection with specific acquisition functions should be obtained from the Naval Publications and Forms Center or as directed by the contracting officer.)

6.2.2.1 The data requirements of 6.2.2 and any task in sections 3, 4, or 5 of this specification required to be performed to meet a data requirement may be waived by the contracting/acquisition activity upon certification by the offeror that identical data were submitted by the offeror and accepted by the Government under a previous contract for identical item acquired to this specification. This does not apply to specific data which may be required for each contract regardless of whether an identical item has been supplied previously (for example, test reports).

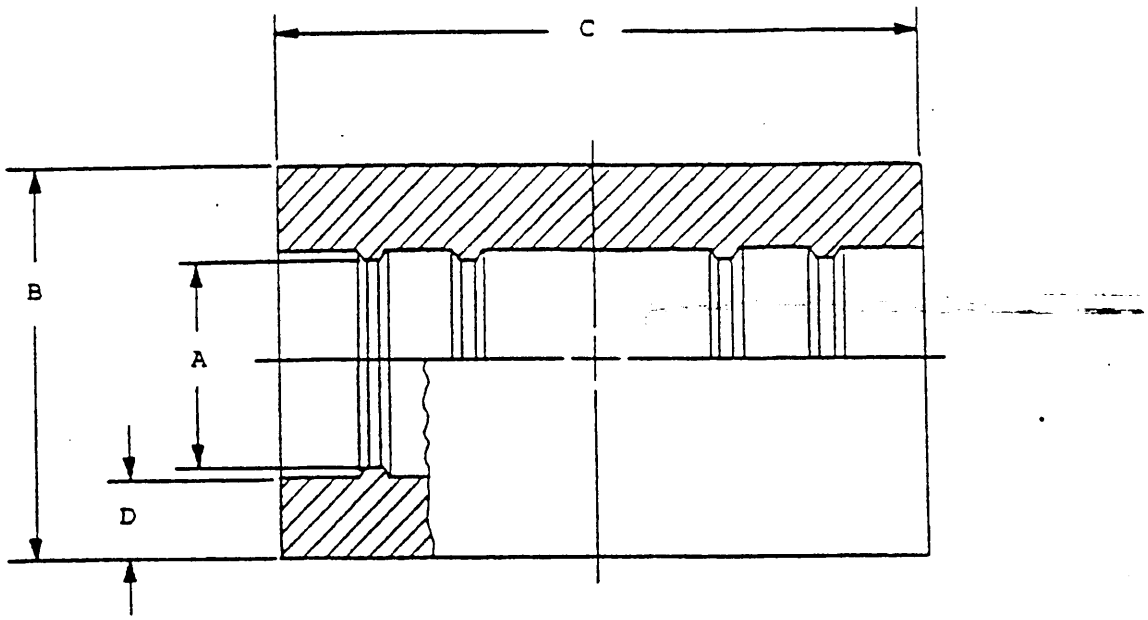
6.3 First article. When a first article inspection is required, the items should be a first article sample. The first article should consist of the units specified in table V. The contracting officer should include specific instructions in acquisition documents regarding arrangements for examinations, approval of first article test results and disposition of first articles. Invitations for bids should provide that the Government reserves the right to waive the requirement for samples for first article inspection to those bidders offering a product which has been previously acquired or tested by the Government, and that bidders offering such products, who wish to rely on such production or test, must furnish evidence with the bid that prior Government approval is presently appropriate for the pending contract.

6.4 Subject term (key word) listing.

Composite coupling
Heat recoverable material
Monolithic coupling

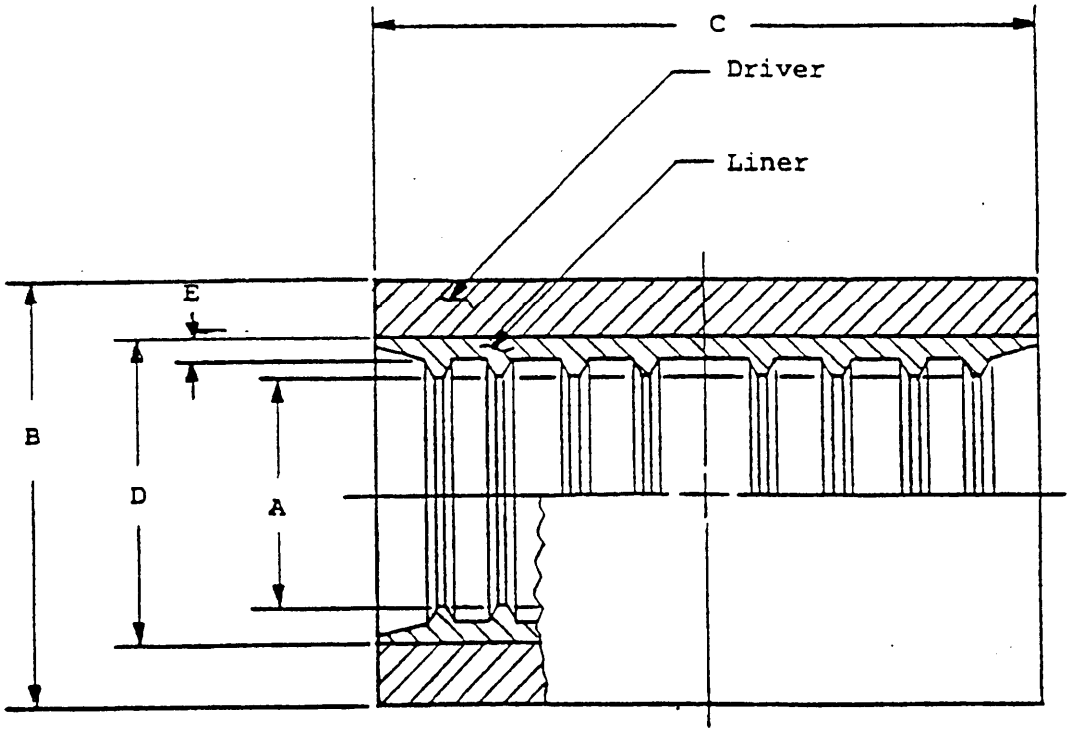
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(Project 4730-N801)

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SH 131909

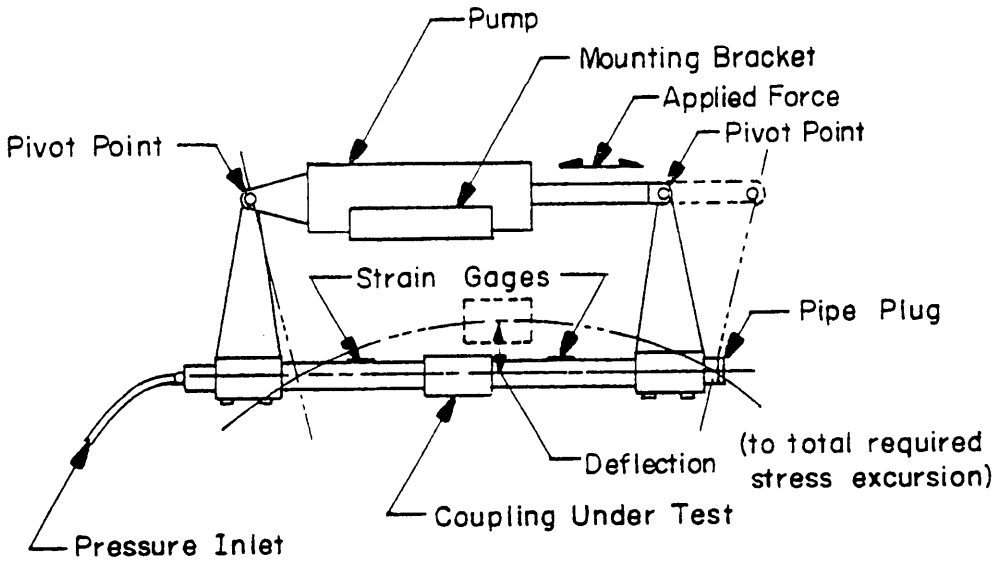
FIGURE 1. Type I coupling (monolithic).



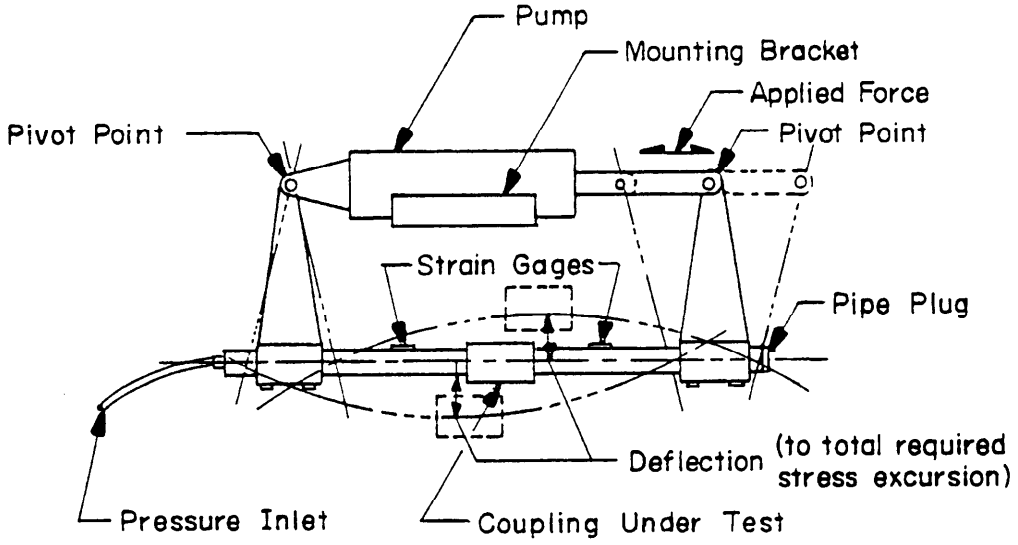
SH 131910

FIGURE 2. Type II coupling (composite).

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Method 1. Deflection from neutral to positive to neutral.



Method 2. Deflection from positive to negative to positive.

SH 131911

FIGURE 3. Planar flexure methods (either method may be used).

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STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

(See Instructions – Reverse Side)

1. DOCUMENT NUMBER	2. DOCUMENT TITLE
MIL-C-24706(SH)	COUPLING, PIPE, HEAT-RECOVERABLE

3a. NAME OF SUBMITTING ORGANIZATION	4. TYPE OF ORGANIZATION *(Mark one)*
	☐ VENDOR ☐ USER ☐ MANUFACTURER ☐ OTHER *(Specify)*:
b. ADDRESS *(Street, City, State, ZIP Code)*	

5. PROBLEM AREAS

a. Paragraph Number and Wording:

b. Recommended Wording:

c. Reason/Rationale for Recommendation:

6. REMARKS

7a. NAME OF SUBMITTER *(Last, First, MI)* – Optional	b. WORK TELEPHONE NUMBER *(Include Area Code)* – Optional
c. MAILING ADDRESS *(Street, City, State, ZIP Code)* – Optional	8. DATE OF SUBMISSION *(YYMMDD)*

(TO DETACH THIS FORM, CUT ALONG THIS LINE.)

DD FORM 82 MAR 1426 PREVIOUS EDITION IS OBSOLETE.

Source: https://assist.dla.mil -- Downloaded: 2026-10-20T01:11Z
Check the source to verify that this is the current version before use.